Unit – III

Cloud Computing
(Course Code: 4360709)

Cloud Service Model & service providers

A. V. PAREKH TECHNICAL INSTITUTE, RAJKOT

Unit-3: Cloud Computing

CO3 - c) To simulate the Service Model with reference to Cloud Computing

3.1.1 Cloud Service Model

- SaaS:
- Brief Introduction
- Advantages and Disadvantages
- PaaS:
- Brief Introduction
- Advantages and Disadvantages
- IaaS:
- Brief Introduction
- Advantages andDisadvantages

3.2.1 Difference between SaaS, PaaS, IaaS

3.3.1 Service provider:

- Amazon web service
- Microsoft Azure
- Google cloud platform
- IBM cloud service
- VMWAre cloud
- Oracle cloud
- Red hat

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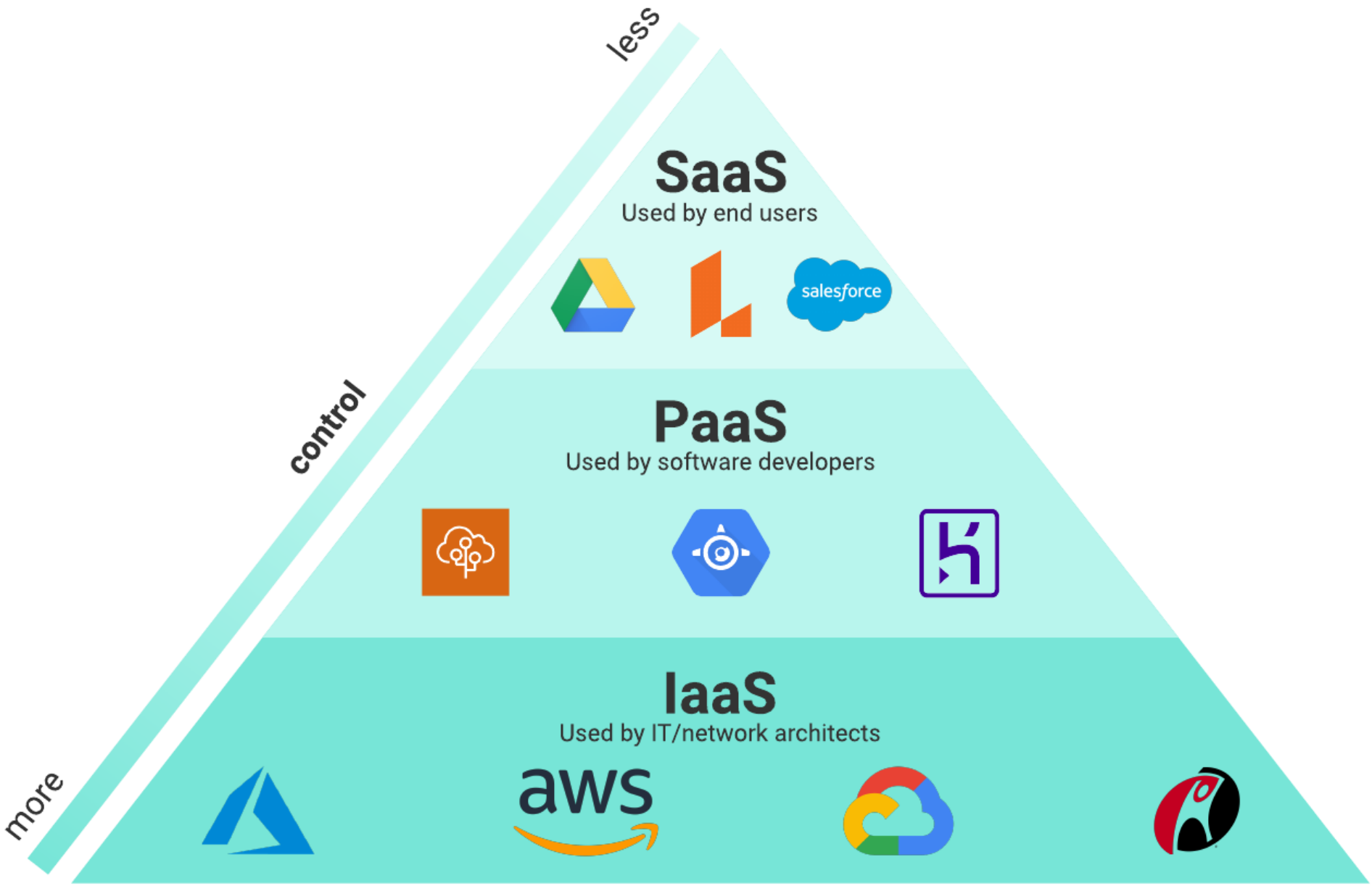

less
control
more
SaaS
Used by end users
salesforce
PaaS
Used by software developers
IaaS
Used by IT/network architects
aws

SaaS

- Cloud application services are the most well-known of the cloud service models.
- The software is hosted, packaged, and delivered by a third party through the Internet (typically on a browser-based interface). By delivering the software application over the Internet, enterprises can offload the costs of management and maintenance to the vendor(s).
- Popular SaaS options include email and customer relationship management software.

Service Models

Software as a Service (SaaS)

- SaaS delivers software applications over the internet on a subscription basis.
- Users can access and use applications directly through a web browser or APIs.
- The provider hosts and manages the underlying infrastructure, application, and data.
- Users can typically customize certain aspects of the application to fit their needs.
- Examples of SaaS include Salesforce, Microsoft Office 365, and Google Workspace.

SaaS Advantages

- **Lower up-front cost :**SaaS is generally subscription-based and has no up-front licence fees resulting in lower initial costs. The SaaS provider manages the IT infrastructure that is running the software, which brings down fees for hardware and software maintenance.
- **Quick set up and deployment :** SaaS application is already installed and configured in the cloud. This minimises common delays resulting from often lengthy traditional software deployment.
- **Easy upgrades :** The SaaS providers deal with hardware and software updates, deploying upgrades centrally to the hosted applications and removing this workload and responsibility from you.
- **Accessibility :** All you need to access a SaaS application is a browser and an internet connection. This is generally available on a wide range of devices and from anywhere in the world, making SaaS more accessible than traditional business software installation.
- **Scalability :** SaaS providers generally offer many subscription options and flexibility to change subscriptions as and when needed, eg when your business grows, or more users need to access the service.

SaaS Disadvantages

- **Lack of control :**In-house software application gives businesses a higher degree of control than hosted solutions where control resides with a third party. Typically everyone has to use the latest version of the software application and cannot defer upgrades or changes in the features.
- **Security and data concerns :** Access management and the privacy of sensitive information is a major consideration around cloud and hosted services.
- **Limited range of applications :**While SaaS is becoming more popular, there are still many applications that don't offer a hosted platform.
- **Connectivity requirement :**Since the SaaS model is based on web delivery, if your internet service fails, you will lose access to your software or data
- **Performance :**SaaS may run at somewhat slower speeds than on-premise client or server applications, so it's worth keeping performance in mind your software isn't hosted on a local machine.

Real world

SaaS examples

Google Workspace

Dropbox

MailChimp

Slack

Microsoft Office 365

DocuSign

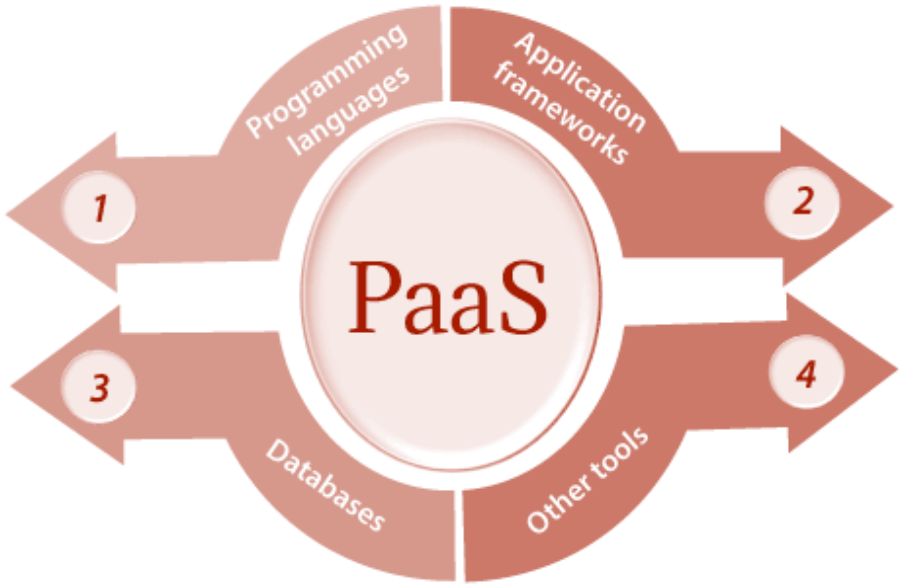
HubSpot

PaaS

- This service layer is primarily geared towards developers and operations professionals.
- Service providers rent out cloud-based platforms for users to develop and deliver applications. In other words, PaaS provides a framework that makes it easier and more efficient to build, customize, and deploy applications.

Platform as a Service (PaaS)

PaaS offers a platform for developing, testing, and deploying applications.
Users can focus on application development without worrying about infrastructure management.
PaaS providers manage the underlying infrastructure, including servers, storage, and networking.
Developers can leverage pre-configured environments, development frameworks, and deployment tools.
Examples of PaaS providers include Heroku, Google App Engine, and AWS Elastic Beanstalk.



PaaS Advantages

1. Multi-language development Having a PaaS means you can get multiple devs to work on it at the same time. Some devs may know programming languages that others don't, but it also means those very devs can work on the same project without having to enlist the help of another server.

2. Rich development environments PaaS come with their own infrastructure, meaning the PaaS solution you choose can come with everything you need to develop apps, as well as make instantaneous changes to them ready for deployment.

3. Can be used by all You don't necessarily need an app developer or dedicated IT personnel to build an app solution on your public PaaS.

4. Cheaper to run You won't have to build any infrastructure for your dev environment, meaning you already cut costs before you begin developing.

5. They're scalable Add more capacity during peak periods, or scale down resources during quiet ones. This can be done quickly with your provider, so that you experience little to no downtime.

6. Speed This is not only isolated to the speed of development time, but also marketing your business. With faster dev times, you can deploy and run apps much faster than with a physical structure that requires provisioning yourself. This means you can develop and update apps as and when needed.

PaaS Disadvantages

1. Less control over infrastructure : A PaaS's infrastructure is managed by your cloud hosting provider. However, if you require something a bit more bespoke, you may need to consider what can (and can't) be tailored. This is why some businesses prefer private clouds, or an IaaS that can allow you to build it yourself.

2. Can be more expensive than an IaaS : Every public cloud provider has a service fee, including IaaS. However, because a PaaS's infrastructure is handled by the cloud provider, there may be additional or higher fees as a result. By having an IaaS, you can build your own infrastructure and integrate the tools you need without having to incur additional fees, or moving to a different PaaS provider altogether.

3. Cloud vendor lock-in : Relying on any public cloud solution, including a PaaS, may result in vendor-lock in and over-dependency. It's important to back-up your work regularly, and ensure if the worst happens with your PaaS that you can easily move to another provider or invest in a private PaaS instead.

4. Security concerns : Any type of public cloud, including PaaS, is managed by the provider – including the security of the cloud solution. However, while PaaS providers take necessary precautions to secure their platform, businesses will need to have robust security measures around the apps they build. Plus, there's the added issue of being on a public cloud solution to begin with, which is shared by other users on their own VMs. If you work with sensitive data or solutions for your business, then you'll need to consider the security risks associated with this.

5. Integration issues : Leading back to our vendor lock-in issue somewhat, usually, running multiple PaaS or merging them together can present some issues. Switching PaaS providers isn't as simple as it sounds either, as the infrastructure has to be able to support the languages used to develop your apps, as well as provide the service needed to run your business specifically.

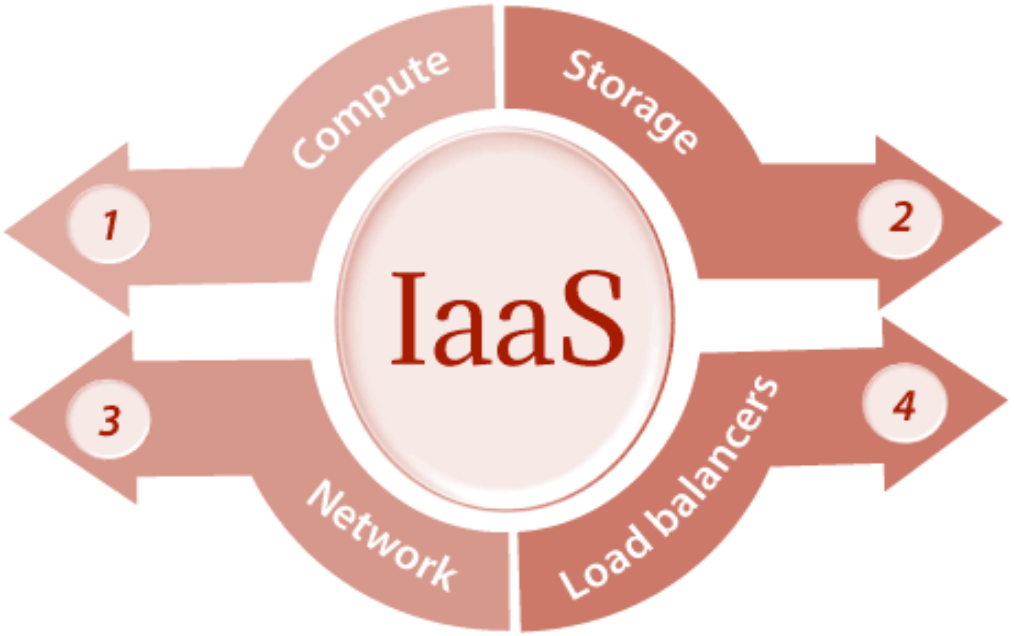
IaaS

- IaaS gives users access to storage, networking, servers, and other computing resources via the cloud. While the user is still responsible for managing their applications, data, middleware, etc., IaaS provides automated and scalable environments that provide a high degree of control and flexibility for the user.
- For example, many businesses use IaaS to support workload spikes during busy seasons (like holidays).
- Popular IaaS providers include:
- Amazon Web Services (AWS)
- Microsoft Azure
- Google Compute Engine (GCE), the IaaS component of Google Cloud Platform (GCP)

Infrastructure as a Service (IaaS)

IaaS provides virtualized computing resources over the internet. Users have control over the operating systems, storage, and networking components.

They can provision and manage virtual machines (VMs), storage, and networks according to their requirements. Examples of IaaS providers include AWS EC2, Azure Virtual Machines, and Google Compute Engine.



IaaS Advantages

1. Cost reduction : When searching for new business solutions, the service that delivers the most for the best price will always be in favour. By eliminating the need for businesses to buy and maintain their own IT hardware, IaaS can significantly reduce capital expenditure. Plus, with pay-as-you-go billing, you only pay for what you use, reducing overall costs.

2. Flexibility : Due to the lack of vendor management, IaaS offers the most flexibility out of the IaaS, PaaS and SaaS cloud computing services. You're free to manage your own OS, software, middleware and applications, allowing you to have full control over your business outside of resource provision.

3. Increased scalability : Since you're renting resources as needed, you can quickly and easily scale up and down to accommodate different workloads. This will prevent you from being overwhelmed during busy periods and overpaying for extra resources at quieter times. This is especially useful for fast-growing businesses, as you'll be able to scale your resources to support your business at all times.

4. Increased performance : Thanks to this scalability, you'll have the performance you need to take on any spike in workload. You'll be able to deliver resources faster than ever before to support your teams and grow your business.

5. Increased stability : It's your provider's responsibility to perform hardware maintenance, troubleshoot issues and ensure that your infrastructure meets service-level agreements (SLAs). This gives you an incredibly reliable and stable IT infrastructure that you don't even have to manage yourself.

6. Improved security : Thanks to these SLAs, your provider is committed to ensuring the security of your infrastructure against potential attacks and breaches. Often, the security they provide can be better than the measures you implement in-house. Plus, if you need extra privacy and security, you can opt for private cloud IaaS to isolate your infrastructure from other organisations.

7. More control: IaaS providers manage your IT infrastructure, but you retain control over everything else. This means you can install your own software, OS and applications and scale your resources up and down whenever you wish.

IaaS Disadvantages

1. More training and expertise required

Less vendor management can be positive or negative depending on your needs. Although IaaS gives you more control, you'll also have more responsibility for managing your software and applications. This can be time-consuming, especially if you need to train your staff to deal with the new systems.

2. Process changes

Switching to IaaS could cause difficulties with some of your legacy systems if they don't integrate well with the new infrastructure. These systems may need to be updated before you move over to IaaS, which can be difficult and time-consuming.

3. Possible security risks

Although IaaS can improve security if your provider implements robust security measures (such as firewalls, encryption and multi-factor authentication), it can also compromise security in some cases. In particular, if you use a multi-tenant environment like a public cloud, you could be vulnerable to cyber attacks and a lack of data privacy. Also, since you're still in control of your software, OS and apps, you'll ultimately be responsible for mitigating security threats in these areas.

4. Vendor lock-in

Once you've switched to one IaaS provider, you may find it difficult to move to another one as this could cause issues for your infrastructure. This can make IaaS less flexible than it appears, as you may be over-dependent on one provider.

Difference between IaaS PaaS & SaaS

Basis Of	IAAS	PAAS	SAAS
Stands for	Infrastructure as a service.	Platform as a service.	Software as a service.
Uses	IAAS is used by network architects.	PAAS is used by developers.	SAAS is used by the end user.
Access	IAAS gives access to the resources like virtual machines and virtual storage.	PAAS gives access to run time environment to deployment and development tools for application.	SAAS gives access to the end user.
Model	It is a service model that provides virtualized computing resources over the internet.	It is a cloud computing model that delivers tools that are used for the development of applications.	It is a service model in cloud computing that hosts software to make it available to clients.
Technical understanding.	It requires technical knowledge.	Some knowledge is required for the basic setup.	There is no requirement about technicalities company handles everything.

Difference between IaaS PaaS & SaaS

Basis Of	IAAS	PAAS	SAAS
Popularity	It is popular among developers and researchers.	It is popular among developers who focus on the development of apps and scripts.	It is popular among consumers and companies, such as file sharing, email, and networking.
Percentage rise	It has around a 12% increment.	It has around 32% increment.	It has about a 27 % rise in the cloud computing model.
Usage	Used by the skilled developer to develop unique applications.	Used by mid-level developers to build applications.	Used among the users of entertainment.
Cloud services.	Amazon Web Services, sun, vCloud Express.	Facebook, and Google search engine.	MS Office web, Facebook and Google Apps.
Enterprise services.	AWS virtual private cloud.	Microsoft Azure.	IBM cloud analysis.
Outsourced cloud services.	Salesforce	Force.com, Gigaspaces.	AWS, Terremark
User Controls	Operating System, Runtime, Middleware, and Application data	Data of the application	Nothing
Others	It is highly scalable and flexible.	It is highly scalable to suit the different businesses according to resources.	It is highly scalable to suit the small, mid and enterprise level business

Service Providers

- Amazon web service
- Microsoft Azure
- Google cloud platform
- IBM cloud service
- VMWAre cloud
- Oracle cloud
- Red hat

1. Amazon Web Services

- Amazon is one of the largest cloud computing companies, which provides cloud storage and compute power for your data.
- What is AWS?
  - AWS stands for Amazon Web Services. It is a cloud computing platform where you can store and process your data on the cloud server. AWS offers several different services that you can use to host your applications, websites or databases on its servers.

Amazon Web Service

- Amazon Web Services, Inc. is a subsidiary of Amazon that provides on-demand cloud computing platforms and APIs to individuals, companies, and governments, on a metered, pay-as-you-go basis. Clients will often use this in combination with auto scaling

- Products and services including computing, storage, networking, database, analytics, application services, deployment, management, machine learning,mobile, developer tools, RobOps and tools for the Internet of Things.
- The most popular include Amazon Elastic Compute Cloud (EC2), Amazon Simple Storage Service (Amazon S3), Amazon Connect, and AWS Lambda (a serverless function that can perform arbitrary code written in any language that can be configured to be triggered by hundreds of events, including http calls)

2. Google Cloud

- Google has made it easy to use the data by offering a wide range of services.
- Google's Cloud is a great solution for any type of business.
- The best thing about Google's Cloud is that it offers an array of services, which can be accessed from anywhere and anytime. Google is a trusted name in the world of technology.
- They have been around for many years and they have always remained on top.
- They also provide data security and data backup. With the number of customers increasing day by day, Google is still the most preferred service provider.

Google Cloud

- Google Cloud Platform (GCP), offered by Google, is a suite of cloud computing services that provides a series of modular cloud services including computing, data storage, data analytics, and machine learning, alongside a set of management tools.
- It runs on the same infrastructure that Google uses internally for its end-user products, such as Google Search, Gmail, and Google Docs, according to Verma, et.al.
- Registration requires a credit card or bank account details.
- Google Cloud, which includes the Google Cloud Platform public cloud infrastructure, as well as Google Workspace (G Suite), enterprise versions of Android and ChromeOS, and application programming interfaces (APIs) for machine learning and enterprise mapping services.

3. Microsoft Azure

- Microsoft has launched a new cloud platform called ‘Microsoft Azure’ and I am sure it will change the landscape of the cloud industry.
- It is known that the Azure is the next generation of cloud services, which is being offered by Microsoft.
- The most prominent thing about it is that it offers the features of both private and public clouds.
- This will be very useful for enterprises as it will allow them to have the flexibility to choose which cloud service they wish to use.

- Microsoft Azure, often referred to as Azure (/ˈæʒər, ˈeɪʒər/ AZH-ər, AY-zhər, UK also /ˈæzjʊər, ˈeɪzjʊər/ AZ-ure, AY-zure),[2][3][4] is a cloud computing platform run by Microsoft.
- It offers access, management, and the development of applications and services through global data centers.
- It also provides a range of capabilities, including software as a service (SaaS), platform as a service (PaaS), and infrastructure as a service (IaaS).
- Microsoft Azure supports many programming languages, tools, and frameworks, including Microsoft-specific and third-party software and systems.

4. Oracle

- Oracle is the topmost cloud provider that provides the best quality services at affordable rates. It has been delivering excellent cloud solutions to its clients for several years now. It is the most reliable platform where businesses can grow and flourish.
- It is one of the most trusted cloud platforms that offer high-end solutions at reasonable rates. It has been providing different services like database, development, testing, and deployment to different industries.

- Oracle Cloud provides Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), and Data as a Service (DaaS).
- These services are used to build, deploy, integrate, and extend applications in the cloud.
- This platform supports numerous open standards (SQL, HTML5, REST, etc.), open-source applications (Kubernetes, Spark, Hadoop, Kafka, MySQL, Terraform, etc.), and a variety of programming languages, databases, tools, and frameworks including Oracle-specific, Open Source, and third-party software and systems.

ORACLE
Cloud

5. IBM

- IBM is the third largest cloud computing company in the world and it also provides cloud computing services for various clients.
- IBM cloud is a flexible, scalable and secure solution to manage any business. The company provides many services such as Big data, Software, Storage, Analytics, Security, etc. which make the company a big name in the IT industry.

IBM CLOUD: AN OVERVIEW

CLOUDWITHEASE

A Rendezvous with Cloud Technologies

IBM **Cloud**

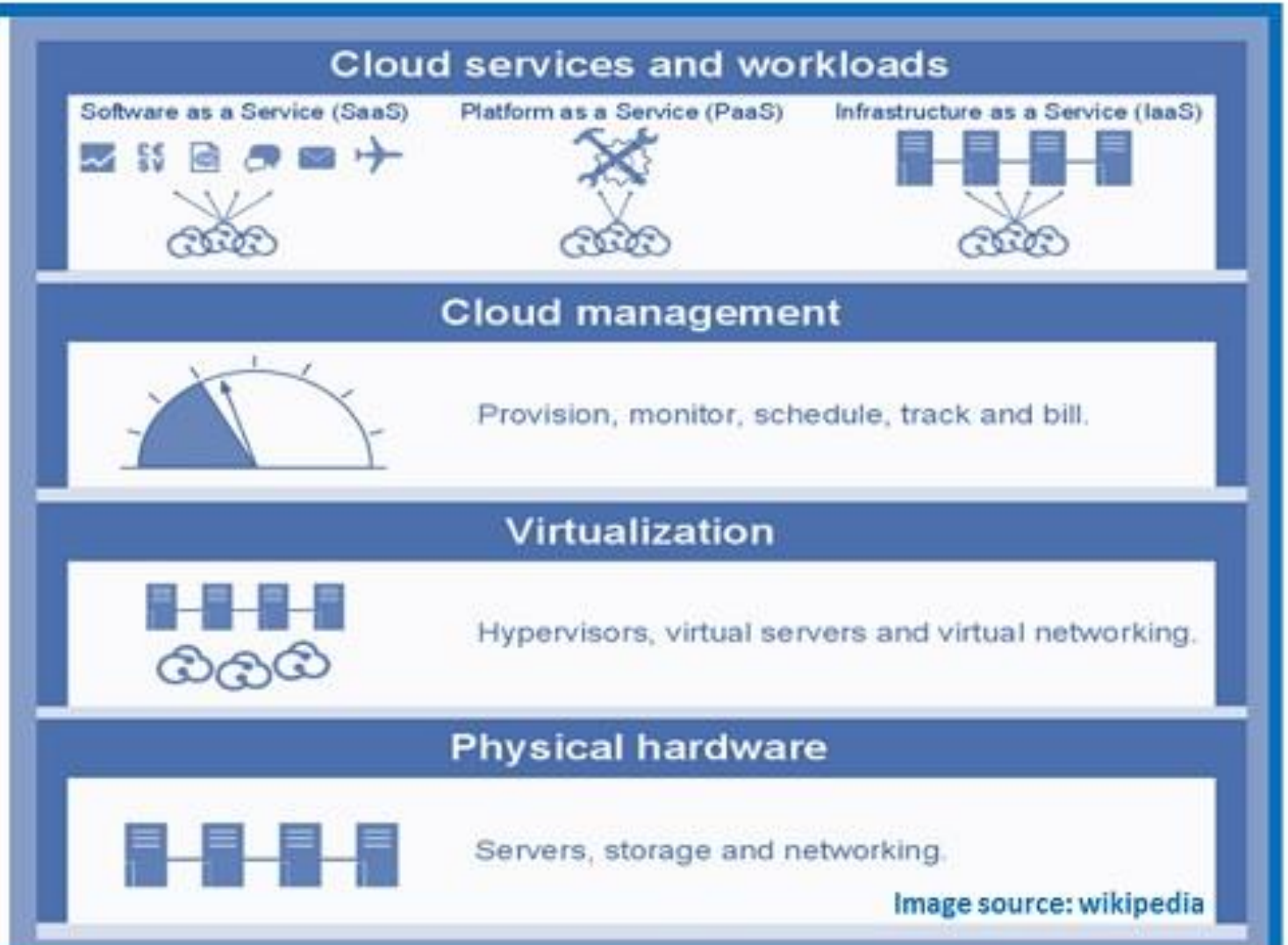


Image source: wikipedia

6. Red Hat

- This company offers cloud services and it is one of the fastest growing cloud companies.
- When it comes to the Red Hat cloud, it provides you the best security as compared to any other tool that is available. The reason behind this is that, the Red Hat cloud has an advanced firewall system. It will help to keep your device safe from the malicious threats.
- Another thing is that, the Red Hat cloud provides you with the most efficient storage capacity. If you want to store your important files, then you can do it with the Red Hat cloud.
- It provides you the most convenient method to access your files. It will help you to save your time.

- Red Hat helps from the initial planning stages to prepare your team to create production-ready environments and ensures your success from day one and beyond.
- With Red Hat Cloud Services, businesses can accelerate innovation, reduce costs, and improve efficiency, gaining a competitive advantage.
- **Optimized value :** Offloading cloud services to a trusted partner can provide predictable, reduced costs.
- **Focus on business priorities :** Prioritize focus on the developers responsible for the applications that drive business.
- **Plugging resource gaps :**Skill and talent gaps impact application delivery when distributed services, hybrid cloud, and cybersecurity are maintained in-house.

7. VMware

- It's a software and technology service that allows customers to deploy, manage and operate VMware-based virtualized workloads over the internet, offering elastic, on-demand capacity that can be scaled up or down at any time.
- Cloud services provides the ability to create, deploy and manage applications in a virtual environment and provides a way to build, test and run software.
- It helps you to get rid of the limitations of physical servers and save a lot of money. In a cloud computing environment, you have the freedom to move your applications anywhere on the internet.

- IBM Cloud for VMware Solutions enables you to seamlessly migrate and modernize ,
- VMware workloads to the cloud, allowing you to leverage your existing investments for a consistent VMware experience—retaining the same level of access, security and control.
- **209% return on investment**
- **USD 1.2 million savings from increased operational efficiency**
- **Protect and future-proof VMware investments**

Google Cloud Platform	Amazon Web Services[41]	Microsoft Azure[42]	Oracle Cloud[43]
Google Compute Engine	Amazon EC2	Azure Virtual Machines	Oracle Cloud Infra OCI
Google App Engine	AWS Elastic Beanstalk	Azure App Services	Oracle Application Container
Google Kubernetes Engine	Amazon Elastic Kubernetes Service	Azure Kubernetes Service	Oracle Kubernetes Service
Google Cloud Bigtable	Amazon DynamoDB	Azure Cosmos DB	Oracle NoSQL Database
Google BigQuery	Amazon Redshift	Azure Synapse Analytics	Oracle Autonomous Data Warehouse
Google Cloud Functions	AWS Lambda	Azure Functions	Oracle Cloud Fn
Google Cloud Datastore	Amazon DynamoDB	Azure Cosmos DB	Oracle NoSQL Database
Google Cloud Storage	Amazon S3	Azure Blob Storage	Oracle Cloud Storage OCI

THANK YOU